



Pulse

Content Analytics & Operations · Media House

PRODUCT REQUIREMENTS DOCUMENT

Pulse — the operating system for a multi-channel media house

One workspace to plan content, track editing, measure performance, and rank the team — across every brand channel and social platform.

DOCUMENT

PRD v2.0 — Current build

PRODUCT

Pulse

ORGANIZATION

Media House (DF Foods)

STATUS

Built & running

LAST UPDATED

21 Jul 2026

4

Brand channels

3

Platforms
(extensible)

10

Live accounts

1

Shared editing
team

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01

Overview & Goals

Media House runs several health-and-food brands (Doctor Farmer, My Health School, My Health Summit, Fofitos) across Instagram, Facebook and YouTube. Pulse replaces the scattered spreadsheets and group chats with a single system that covers the whole content lifecycle — from planning a post, through editing, to measuring how it performed and who did the work.

The problem

- Content lives in a manual entry sheet; metrics are copied by hand and quickly go stale.
- No shared view of what each editor is working on or where a post is in the editing pipeline.
- Performance can't be compared across channels or platforms, and there's no objective way to rank editors.
- Adding a new brand or platform means rebuilding the sheet.

Goals

Single source of truth

Every post, metric, editor and task lives in one database, scoped to the Media House organization.

Multi-platform by design

Channels × platforms modelled as first-class “accounts”, so new brands and networks are configuration, not code.

Operational visibility

See the editing pipeline and each editor's workload at a glance; tasks appear automatically when work is assigned.

Objective performance

Transparent, formula-based scoring ranks both content performance and work discipline.

Non-goals (this version)

- Automatic metric import from platform APIs — planned; metrics are entered manually for now.
- Platform-specific metric sets (watch-time, subscribers) — a shared metric set is used across platforms.
- Public content publishing/scheduling — Pulse tracks content, it does not post it.

02

Users & Roles

All users belong to the Media House organization. Access is governed by a per-user role. Editors (the people who edit content) are a separate concept from login users — see §3.

ROLE	CAN DO	CANNOT DO
Admin	Everything: manage users, channels, platforms, taxonomy, editors; create & edit content; run tasks.	—
Editor	Create & edit posts and tasks; advance editing stages; view all analytics.	Manage users, taxonomy, editors, or platform setup.
Viewer	Read-only access to dashboards, metrics and leaderboards.	Any create / edit / delete action.

BOOTSTRAP ADMIN

A single environment-configured admin is guaranteed access on every startup and can never be locked out of management — the safety net for the whole organization.

03

Core Concepts & Data Model

Pulse is built on a small set of entities. Understanding these makes every feature obvious.

ENTITY	WHAT IT IS	SCOPE	KEY FIELDS
Organization	The “Media House” — the top-level container for everything.	Global (one)	name
Channel	A brand account (Doctor Farmer, Fofitos...). Holds content & taxonomy.	Org	name, org
Platform	A social network — Instagram, Facebook, YouTube. Extensible.	Global reference	key, name, order
Account	One channel on one platform (“Doctor Farmer on YouTube”). The real data source.	Channel × Platform	handle, external_id, token*
Editor (team member)	A person who edits content. Shared across all channels.	Org	name, designation, photo
Post	A single piece of content, tied to a channel + platform.	Channel + Platform	see §7
Task	A unit of work assigned to an editor. Manual or auto-created from a post.	Org	title, assignee, due, status, post link
Taxonomy	Content Pillars, Content Types, Formats, Audience Avatars.	Channel	see §7

* **token** is reserved for future API auto-sync and is never exposed to the browser.

THE CENTRAL IDEA

Content is a grid of **Channel × Platform**. Each filled cell is an **Account** — the thing that has posts and analytics. This is what makes Pulse scale to any number of brands and networks.

04

Channels & Platforms

A channel can exist on many platforms; not every channel is on every platform. That presence is stored explicitly as the account grid.

CHANNEL	INSTAGRAM	FACEBOOK	YOUTUBE
Doctor Farmer	✓	✓	✓
My Health School	✓	✓	✓
My Health Summit	✓	✓	—
Fofitos	✓	✓	—

FR-4.1

Add a channel — an admin can create a new brand channel; it is seeded with the standard taxonomy and joins the org.

FR-4.2

Enable a platform — an admin can turn a platform on/off for a channel, creating or removing its account. A platform with existing posts cannot be removed.

FR-4.3

Extensible platforms — adding a new platform (e.g. LinkedIn, TikTok) is a single reference row plus per-channel enablement. No schema change.

Pending

A self-service Channels & Platforms management screen (FR-4.1/4.2 via UI) is the one remaining piece; today channels are provisioned by the team.

05

Dashboard

Platform-first analytics. Pick a channel scope at the top, then open a platform to see its numbers.

FR-5.1

Channel selector — “All Channels” (aggregated) or a single brand, at the top of the dashboard. Replaces the old top-bar workspace switcher.

FR-5.2	Platform cards — one card per platform the selected channel is on, each showing a post-count and view summary. Only real accounts appear.
FR-5.3	Drill-in — clicking a platform card expands its full analytics in place: no navigation.
FR-5.4	Analytics blocks — KPI tiles (Total Posts, Views, Accounts Reached, Engagement Rate, with period-over-period deltas), Views by Format (reels vs carousels), Top Performers (best reel & carousel by Performance Score), Views by Pillar, Views by Audience Avatar, Content Mix, and Engagement totals.
FR-5.5	Date range — presets and a custom range filter all analytics; only Published posts count.

06

Metrics — Editing Progress

A live overview of every piece of content moving through the editing pipeline, across all channels. Distinct from publish status: a post can be scheduled while its edit is still in review.

Editing stages

Not Started → In Progress → In Review → Pending → Completed

FR-6.1	KPI cards — Total / Completed / In Progress / Pending counts for the current filter.
FR-6.2	Progress donut — percentage completed, with a legend counting every stage.
FR-6.3	Pipeline table — each post shows title + content pillar, channel (platform logo only), assigned editor (avatar + name), a stage badge, due date and last-updated time.
FR-6.4	Inline stage change — change any post's stage directly from the board; KPIs and donut update live.
FR-6.5	Filters — by editor and by stage.

Content & Posts

A post is one piece of content on one account. Content is created in bulk and edited one at a time.

Post fields

FIELD	DESCRIPTION	REQUIRED
Channel	The brand account the post belongs to.	Yes
Platform	Instagram / Facebook / YouTube (limited to the channel's accounts).	Yes
Date	Scheduled / published date.	Yes
Title	Post title.	Yes
Link	URL of the published post.	No
Collab	Another channel this post is a collaboration with.	No
Content Pillar	Top-level content theme (e.g. Diabetes, Obesity).	Yes
Content Type	Sub-category within the pillar.	Yes
Format	Named format template within the pillar.	Yes
Post Type	Reel or Carousel.	Yes
Audience Avatar	Target audience persona (powers avatar analytics).	Yes
Assigned Editor	Team member responsible — auto-creates a task (§8).	No
Publish Status	Planned or Published.	Yes
Editing Stage	The §6 pipeline stage.	Default: Not Started

FIELD	DESCRIPTION	REQUIRED
Metrics	Views, Likes, Comments, Shares, Saves, Accounts Reached.	After publishing

Bulk & single entry

FR-7.1	Add Multiple Posts — a spreadsheet-style modal where each row is a post. Add / duplicate / delete rows, then Save All . Per-row cascading: platform follows channel; content type & format follow pillar; each row uses its own channel's taxonomy.
FR-7.2	Edit a post — clicking a post opens a single-post glass modal with all fields (including Link & Collab), plus a delete action.
FR-7.3	Posts database — searchable, filterable (All / Planned / Published / Reels / Carousels) table with CSV export.
FR-7.4	Content taxonomy — Pillars, Audience Avatars, and per-pillar Content Types & Formats are managed in Settings (admin-only add/delete; deletion blocked while in use).

08

Task Management

A kanban board (To Do → In Progress → Done) for the team's daily work, in its own "Team" section.

FR-8.1	Manual tasks — create a standalone task with a title, details, assignee, due date and priority, unlinked to any post.
FR-8.2	Automatic tasks — assigning an editor to a post auto-creates a linked task for that editor (title, channel, due date carried over). Editing the post keeps the task in sync. Auto-tasks show a  Post badge.
FR-8.3	Board operations — assign, set priority, move status (drag-equivalent buttons), edit and delete. Filter by team member.

TWO WORK VIEWS, ONE PURPOSE

Task Management is the team's to-do board; **Metrics** (§6) is the content pipeline. Complementary — tasks are people-centric, the pipeline is content-centric.

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Leaderboard

Two ways to lead, on two tabs — the same team ranked on two independent axes.



Social Media Leaders

Ranked on the performance of the Reels & Carousels each editor produced, aggregated across every channel. Uses weighted-engagement points (see §11), with a This-Month / All-Time toggle and a featured #1 champion card.



Media House Leaders

Ranked on work discipline — **task completion rate**, with total tasks completed as the tiebreaker so consistency beats a lucky single task. Its own “Top Performer · Work” champion card.

FR-9.1

Featured champion — the #1 editor's photo, role and headline stats are shown prominently and update automatically with the ranking.

FR-9.2

Medals — top three get gold / silver / bronze laurels; the rest are numbered; editors with no qualifying work are listed but unranked.

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Insights, Reports & Settings

Insights

Rule-based observations computed automatically from published data — top pillar, format that wins on saves, under-

Reports

Export and share performance summaries (CSV today; richer exports on the roadmap).

served audience — each with a recommendation. No AI writing; fully explainable.

Settings

Content taxonomy management, the editing team (add/edit/remove editors with profile photos), and scoring reference.

User Management

Admin-only: create users, assign roles, grant or revoke access, deactivate accounts — with last-admin and self-lockout guards.

11 Scoring Formulas

Every ranking is a transparent formula — no black boxes.

Performance Score (content)

Each engagement metric is first converted to a rate as a percentage of reach, then weighted. Reach of zero yields a score of zero.

$$\text{Rate} = (\text{metric} \div \text{reach}) \times 100$$

Performance Score =

$$(\text{View Rate} \times 20) + (\text{Like Rate} \times 10) + (\text{Comment Rate} \times 15) \\ + (\text{Share Rate} \times 30) + (\text{Save Rate} \times 25)$$

Weights emphasize the strongest quality signals — shares and saves. Used for the dashboard's "best performer" cards.

Social Media Leaders — points

$$\text{Points} = \text{Likes} + (2 \times \text{Comments}) + (3 \times \text{Shares}) + (3 \times \text{Saves})$$

Summed across an editor's published Reels & Carousels for the selected period.

Media House Leaders — Work Score

Completion Rate = tasks completed ÷ tasks assigned
Rank by Completion Rate, tiebreak by tasks completed

On-time delivery and consistency are captured in the data model and reserved for a future weighting.

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Non-Functional Requirements

AREA	REQUIREMENT
Performance	Warm, persistent database connection pool with heartbeat (avoids ~2.4s cold-connect to the remote DB); stale-while-revalidate client cache so revisited pages render instantly. Post list responds in ~0.25s.
Security	httpOnly JWT session cookie, bcrypt-hashed passwords, per-request role resolution. Secrets (DB, JWT, admin) live in server env only, never in the client or repo.
Responsive	Full layouts at 992 / 768 / 576 / 350 px; off-canvas drawer navigation on mobile; horizontally-scrolling tables never break the page.
Theming	Purple glassmorphism design system; light & dark themes (dark default) via CSS tokens.
Accessibility	Readable contrast in both themes; keyboard-operable modals (Escape / backdrop close); respects reduced-motion.
Data integrity	Additive, backed-up migrations; foreign-key guards prevent deleting taxonomy or platforms still in use.

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Architecture & Tech Stack

LAYER	TECHNOLOGY	NOTES
Frontend	Vite · React 19 · TypeScript · React Router	Single-page app; custom purple glass design system in CSS.
Backend	Node · Express	REST API under <code>/api</code> ; JWT auth; org & workspace resolution middleware.
Database	Supabase Postgres (Sydney)	Used as pure Postgres via <code>pg</code> — no Supabase Auth or RLS. Migrations in SQL.
Auth	JWT (httpOnly cookie) · bcryptjs	DB-backed users; env-configured bootstrap admin.

In production the Express server also serves the built React app, so the whole product runs as **one service on one origin** — eliminating cross-site cookie and CORS complexity. In development the React app runs on its own Vite dev server against the API.

14 Deployment

DEP-1	Single service on Render (Singapore region, closest to the Sydney database). One build compiles the React app; the Node server serves it and the API.
DEP-2	Config via a <code>render.yaml</code> blueprint; secrets entered as dashboard environment variables (DB URL, JWT secret, admin credentials).
DEP-3	Source on GitHub; auto-deploys on push.
DEP-4	Database is the managed Supabase Postgres instance — shared by local and production.

15 Roadmap

ITEM	WHY	PRIORITY
Channels & Platforms management UI	Add channels / toggle platforms without engineering.	Next
Platform API auto-import	Pull metrics automatically (accounts already hold token fields).	High
Platform-specific metrics	Watch-time, subscribers (YouTube); reactions (Facebook).	High
On-time & consistency in Work Score	Reward reliability, not just completion.	Medium
Object storage for images	Full-resolution editor & content imagery.	Medium
Metrics timeline view & notifications	Trend of the pipeline; alerts on due/overdue work.	Medium

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Glossary

TERM	MEANING
Media House	The organization — the single top-level tenant that owns all channels, team and data.
Channel	A brand (Doctor Farmer, Fofitos...). Formerly “workspace”.
Account	A channel on a specific platform — the unit that holds posts and analytics.
Editor	A person who edits content; a shared, org-level team member (not a login role).
Publish Status	Whether a post is Planned or Published — drives analytics inclusion.
Editing Stage	Where a post sits in the production pipeline (Not Started → Completed).

TERM

MEANING

Pillar / Type / Format / Avatar

The four-part content taxonomy that classifies every post.

Pulse — Product Requirements Document v2.0. Prepared for Media House (DF Foods). Reflects the current running build across dashboard, metrics, content, tasks, and leaderboard. Sections marked **Pending** / roadmap are not yet shipped.